

The Limits of Platform Observability

Statements of Reasons on Dating Apps under the Digital Services Act

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Abstract

When online platforms restrict user content or accounts, affected users and external observers often struggle to understand what was decided, why it was decided, and how moderation practices operate on a scale. Article 17 of the Digital Services Act (DSA) seeks to address this opacity by requiring platforms to provide statements of reasons (SoRs) for content moderation decisions and to submit them to a public Transparency Database. This article examines whether these disclosures enable meaningful platform observability in the under-studied context of dating applications, where moderation governs access to intimate, relational, and identity-based forms of social participation. Drawing on the full population of 6,878,730 statements of reasons submitted by *Badoo*, *Bumble*, *Hinge*, *Meetic*, *Tinder*, *Lovoo*, and *OkCupid* over a six-month observation window, we estimate logistic regression models to identify when platforms provide substantive explanations. We find that 79.1% of moderation decisions include such an explanation, with explanation rates ranging from 7.1% to 98.9% across platforms. Rather than by a consistent norm of reason-giving, explanation provision is shaped by platform-level reporting infrastructures, automation pipelines, user-initiated reports, and account termination. Fully automated detection-and-decision pipelines are associated with higher explanation rates, while partial automation and user-initiated reports are associated with lower rates. These findings show that Article 17 DSA disclosures support sector-level observability, but only to a partial extent: they reveal patterns of standardization, platform-specific reporting practices, formalized transparency, and incomplete accountability, while leaving the underlying content, context, and potential discriminatory effects of moderation decisions largely outside view. The article contributes to Information Systems research by analyzing regulatory-imposed transparency databases as socio-technical infrastructures that shape what can, and cannot, be known about platform governance.

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1 Introduction

When online platforms remove content, restrict visibility, suspend services, or terminate accounts, they affect the availability, visibility, and accessibility of user-provided information and may determine whether a user can continue to participate in the service at all (Klonick, 2018). Affected users often struggle to understand what was decided, on what grounds, and through which moderation process (Suzor et al., 2019; Jhaver et al., 2019a). For this reason, the explanation accompanying a moderation decision matters: it allows users to connect the restriction imposed on them to the rules, procedures, and factual circumstances on which the platform relied, and to decide whether there are grounds to contest it (Jhaver et al., 2019b). In practice, affected users may receive little more than a generic notification, leaving them unable to understand why the decision was taken or how the platform’s rules were applied in their case (Suzor et al., 2019). The problem is broader than mere individual redress. It also concerns whether platform enforcement is intelligible to those subject to it and accountable as a form of platform governance. This opacity is especially significant because content moderation is no longer a marginal operational function, but part of the infrastructure through which platforms organize participation and control interaction (Klonick, 2018; Gillespie, 2018; Douek, 2022; Ens et al., 2023). Through terms and conditions, community rules, interface design, automated detection tools, human review procedures, and enforcement practices, platforms shape what users can do, what remains visible, and which forms of conduct become acceptable within the platform environment (Klonick, 2018; Ens et al., 2023).

The Digital Services Act (DSA) addresses this issue by making it a key part of online platform regulation. Under Article 17 of the DSA, hosting service providers must provide affected users with statements of reasons (SoRs) whenever they remove content, or otherwise restrict access to, content, services, or accounts provided by users. Article 24(5) of the DSA then goes further by requiring online platforms to send these statements of reasons to the European Commission’s public Transparency Database. This creates a dual system of transparency: the statements of reasons are intended to help users understand the decisions and, if possible, challenge them. At the same time, including them in the Transparency Database creates a regulatory dataset that enables large-scale, cross-platform examination of content moderation.

Among the various transparency mechanisms identified in the DSA literature, Samuelson and Helberger (2024) distinguish between obligations directed at users; reporting and information-access obligations directed at regulators; data-access rights for vetted researchers; and public database obligations aimed at wider society. This article focuses on transparency obligations directed at users, while relying on the public databases resulting from the latter category as its primary source of data. More precisely, it focuses on

Article 17 DSA, which requires hosting providers to give affected users clear and specific statements of reasons (SoRs) when they remove or otherwise restrict access to user-provided content. Article 24(5) DSA then extends this obligation beyond the individual user by requiring online platforms to submit those SoRs to the European Commission’s (EC) public Transparency Database. The result is a dual transparency mechanism: SoRs are meant to help users understand and potentially challenge moderation decisions, while their publication in the

Transparency Database creates a regulatory dataset through which content moderation can be scrutinized across platforms and at scale.

This promise of scrutiny depends, however, on what the database makes visible. The DSA Transparency Database provides large-scale access to platform-reported data on moderation decisions, including the type of restriction imposed, the source of detection, the use of automated means, the legal or contractual ground relied upon, and the textual explanation provided to the affected user. In principle, this makes it possible to examine content moderation across platforms and at a scale that was previously unavailable. In practice, however, public disclosure does not necessarily deliver substantive accountability (Groesch et al, 2026). Critical transparency scholarship questions the assumption that making information available to users necessarily leads to greater understanding, control, or accountability (Ananny & Crawford, 2018). Transparency can function as a mode of governance built on co-regulation and individual responsibility, where the release of information may substitute for deeper forms of power redistribution and shift the burden of scrutiny from the discloser onto whoever receives the disclosure (Adams, 2018; Stohl et al., 2016; Maroni, 2023). Transparency therefore risks becoming passive and static: a status online platforms can claim by publishing the requested documentation, trading on a pretence of objectivity it cannot deliver, and seldom inviting the kind of external verification that would test its claims (Leerssen, 2024). *Platform observability* has been advanced against this critique as a more practice-oriented alternative (Leerssen, 2024). It is conceptualised as an active and situated practice through which external actors can piece together partial, constructed knowledge out of what online platforms disclose (Rieder & Hofmann, 2020; Papaevangelou & Votta, 2025). Civil society and academia have increasingly taken up this task, critically analysing what providers of intermediary services actually disclose to test whether the transparency obligations under the DSA - and the observability they enable - are meaningful in practice (DSA Civil Society Coordination Group, 2025; Rigotti et al., 2026; Seck & Klotsonis, 2025).

The DSA Transparency Database has already attracted empirical attention. Existing studies have used it to analyze content moderation across major social media platforms, examine inconsistencies in reporting practices, and assess whether statements of reasons deliver the transparency gains promised by the DSA (Drolsbach & Pröllochs, 2023; Kaushal et al., 2024; Papaevangelou & Votta, 2025; Trujillo et al., 2025). These studies show that the DSA database is a major development in platform governance, but also that its usefulness depends on the quality, consistency, and specificity of platform disclosures. This emerging literature, however, has largely focused on large social media platforms and cross-platform moderation patterns, but it has paid much less attention to sector-specific contexts in which moderation decisions govern access to distinctive forms of social participation. Dating applications are one such context.

Dating apps are a compelling case because they are not merely marketplaces for digital services. They are intimate platforms: socio-technical environments through which users seek relationships, sex, recognition, community, and self-presentation (Hutson et al., 2018; Wu & Trottier, 2022). Their governance matters because restrictions on profiles, images, messages, or accounts can affect users' ability to be visible, contactable, and intelligible to others in highly personal and identity-laden settings. The risks associated with dating apps are also unusually layered. Some stem from user conduct, including harassment, abuse, impersonation,

and image-based sexual abuse; others are built into the platform environment itself, including privacy and datafication risks, surveillance-based safety interventions, and design choices that can encode narrow assumptions about gender, sexuality, desirability, and safety (Albury et al., 2017; Bivens & Hoque, 2018; Hutson et al., 2018; Stardust et al., 2023; Wu & Trottier, 2022). Content moderation is therefore presented as a necessary safety measure, but safety interventions on dating apps are not neutral. Existing research shows that platform governance in this sector can be fragmented, opaque, and uneven, and that measures framed as safety can also intensify surveillance or disadvantage already marginalized users (Duguay et al., 2020; Stardust et al., 2023).

This makes dating apps particularly important for studying the limits of platform observability. The same moderation practice that removes abusive content may also remove or suppress sexual self-expression, queer visibility, gender-nonconforming presentation, sex-worker identities, racialized bodies, or other expressions that fall outside a platform’s preferred norms of acceptable intimacy (Narr, 2021; Sommer Degn, 2025). For example, a platform may moderate a profile image as “nudity,” “sexual content,” or a “terms-and-conditions violation,” while the SoRs may not reveal whether the decision concerned exploitative material, consensual sexual self-presentation, queer visual culture, sex-worker identity, or a body read through discriminatory assumptions. Similarly, a user may be suspended after another user’s report, while the public database may reveal only that a user notice triggered the action, not whether the report reflected genuine harm, harassment, malicious flagging, or discriminatory targeting.

Against this background, this article examines the transparency and observability of dating-app content moderation under Article 17 DSA. We focus on seven dating apps operating in the European Union (EU) - *Badoo*, *Bumble*, *Hinge*, *Meetic*, *Tinder*, *Lovoo*, and *OkCupid* - and analyze the SoRs they submit to the DSA Transparency Database. This article asks: *To what extent do the statements of reasons submitted by dating apps to the DSA Transparency Database under Article 17 DSA enable meaningful observability of their content moderation practices?* We answer this question through an empirical analysis of the full population of 6,878,730 row-level SoRs submitted by these seven platforms over a six-month observation window. The analysis focuses on whether platforms provide substantive explanations for moderation decisions, how explanation provision varies across platforms, and whether this variation is associated with automation, user-initiated reports, and the severity of the restriction imposed.

The article makes three contributions. First, it provides the first sector-specific empirical analysis of dating-app content moderation using the DSA Transparency Database. Second, it contributes to Information Systems research by theorizing regulatory transparency databases such as socio-technical infrastructures whose design determines what becomes knowable about platform governance, and what remains structurally invisible. Third, it shows that observability is partial and uneven: the database enables researchers to detect platform-level reporting patterns, automation effects, and systematic differences in explanation provision, while leaving the underlying content, context, and potential discriminatory effects of moderation decisions largely outside view. In doing so, the article shifts the debate from whether platforms formally disclose moderation decisions to whether the resulting data infrastructure supports meaningful scrutiny of how platform governance operates.

The remainder of its article is organized as follows. Section 2 sets out the legal framework governing statements of reasons under Article 17 of DSA and situates the transparency obligation within the Regulation’s broader procedural architecture. Section 3 develops the theoretical argument and derives three testable hypotheses concerning automation, user-initiated reporting, and decision severity. Section 4 describes the DSA Transparency Database, the seven dating platforms analyzed, and the construction of the dependent and explanatory variables. Section 5 presents empirical results, including platform heterogeneity and a battery of robustness checks. Section 6 discusses what the findings reveal about corporate standardization of Article 17 DSA compliance, automation and human oversight, transparency washing, the economics of explanation, and the limits of regulatory data for platform observability. Section 7 concludes.

2 Background information

Article 17 DSA establishes the statement-of-reasons obligation examined in this article. It applies to providers of hosting services - a legal category that includes online platforms such as dating apps (Fosch-Villaronga et al., 2025; Giles et al., 2022) - and requires them to accompany covered content moderation measures taken against a user with a SoR. Overall, the provision performs two related functions: it notifies the affected user that a restriction has been imposed, and it explains the grounds on which that restriction rests (Leerssen et al., 2026). This section situates that obligation within the DSA’s broader procedural architecture before turning to the measures that trigger Article 17 DSA, the information a SoR must contain, and the standard of clarity, precision, and specificity against which such statements must be assessed.

This duty to state reasons must be read against the DSA’s broader regulatory design. The literature converges on the idea that the Regulation introduces a new procedural architecture for platform governance within the EU (Griffin, 2025; Wendel, 2023). Rather than laying down substantive standards determining which content may or may not circulate, the DSA disciplines the way moderation decisions are taken, explained and contested – what Ortolani has described as a “procedure before substance” approach (Ortolani, 2023). Legal scholars accordingly tend to interpret the DSA’s procedural duties as a set of platforms due process guarantees (Juříčková, 2024). Within this architecture, the SoR occupies a pivotal position: precisely because the Regulation largely abstains from dictating substantive outcomes, the protection it affords users hinges on the procedural quality of the decisions taken against them.

Article 17(1) DSA specifies what counts as such a restriction, and on what basis it may be imposed. The obligation applies to four types of measure: (i) restricting the visibility of a specific item of content, whether by removing it, disabling access to it, or demoting it; (ii) suspending, terminating, or otherwise restricting monetary payments; (iii) suspending or terminating the service, in whole or in part; and (iv) suspending or terminating the user’s account. Recital 55 DSA also refers to the practice of shadow banning (i.e., reducing the visibility or discoverability of content or accounts without removing them outright) as an example of such a restriction. Whichever measure is taken, it may rest on either of two grounds: that the content is illegal, meaning that it fails to comply with EU or national law under Article 3(h) DSA - image-based sexual abuse, or that it breaches the provider’s own terms and conditions, which Article 14

DSA requires providers to set out in advance. In the dating app context, the first category may include forms of non-consensual intimate or manipulated material criminalised under Article 5 of Directive (EU) 2024/1385 on violence against women, while the second-typically covers platform rules on matters such as nudity, harassment, solicitation, and the impersonation or misrepresentation of identity (DSA-Database, 2026). Recital 54 DSA clarifies that the obligation applies irrespective of whether the provider relies on illegality or incompliance with its terms and conditions.

Article 17(2) DSA then specifies the conditions of this duty: it covers only cases where the provider knows the affected user's electronic contact details, it applies at the latest from the date the restriction is imposed, regardless of why or how it was imposed, and it does not apply to deceptive high-volume commercial content. When this duty applies. Usually, the provider must know the affected user's electronic contact details, must provide the statement of reasons no later than the moment the restriction is imposed, and the duty does not apply to deceptive high-volume commercial content.

Article 17(3) DSA then enumerates what the SoR must contain. These elements include the type, scope, and duration of the measure; the facts and circumstances underpinning it, including whether it followed a user notice under Article 16 DSA or the provider's own initiative, with the notifying party's identity disclosed only where strictly necessary; whether automated tools played a role in detecting or enforcing the decision; and the legal or contractual basis relied upon. The SoR therefore does more than notify the user of a restriction: it must communicate what measure was adopted, how the decision was reached, and the legal or contractual basis the provider relied on. Art. 17(3)(f) also requires SoR to provide clear and user-friendly information on the redress possibilities available to the affected user. These include, where applicable, internal complaint-handling (Article 20 DSA), out-of-court dispute settlement (Article 21 DSA), or judicial redress. This last requirement matters beyond formal compliance because information on redress is of use only if the SoR given are meaningful. A generic or opaque explanation may notify the user that a restriction has occurred, but it does not equip them to understand, challenge, or correct the decision. This allows the DSA to address the structural imbalance of power between platforms and the people who use them. In this sense, the SoR operates not merely as an information duty but as a precondition for effective redress: the rights of complaint and appeal mechanisms created established by the DSA become exercisable only where the reasons communicated under Article 17 DSA are sufficiently precise and intelligible to be contested (Ortolani, 2023).

Article 17(4) DSA sets the standard for the quality of the information provided. The SoR should be clear and easily comprehensible, and as precise and specific as the circumstances reasonably permit. The standard is demanding because moderation decisions are made at scale. Providers - dating apps among them - process large volumes of profiles, images, and messages, and many decisions are produced through reporting systems, automated decisions, or a combination of both. At that scale, the question is whether each individual SoR can still provide the level of detail to support the rights Article 17 DSA is meant to activate. Without clear interpretative guidance, the obligation risks becoming a mere record of compliance ox-ticking exercise rather than an explanation users can act on (Quintais et al., 2023).

Article 17(4) DSA therefore prevents from becoming a formal checklist and transforms it into a functional requirement: the information provided must be as precise and specific as reasonably possible under the given circumstances and must reasonably allow the recipient to effectively exercise the possibilities for redress referred to in Article 17(3)(f) DSA. The standard is contextual rather than absolute: a severe or complex restriction may require more detail than a routine measure. Yet, a statement that identifies a general category of violation without specifying the conduct at issue may formally comply with Article 17(3) DSA while failing the functional test of Article 17(4) DSA. The user may learn that a restriction occurred, but not what conduct triggered it or why the platform treated that conduct as illegal or incompatible with its terms.

This doctrinal tension underpins the empirical analysis that follows. Article 17 DSA makes reason-giving central to contestation and redress, but the value of that architecture depends on whether platform disclosures include reasons specific enough to serve that function.

3 Hypothesis and Empirical strategy

3.1 From meaningfulness to observable indicators

The preceding section established the legal standard against which Article 17 DSA disclosures should be understood. A meaningful SoR is not simply one that exists, but one that gives the user reasons specific enough to understand the decision and decide whether to contest it or not. The DSA Transparency Database, however, records platform-reported information about moderation decisions without disclosing the underlying content, conduct, user context, or interaction. It therefore does not allow us to assess meaningfulness in the full Article 17(4) sense.

The empirical analysis makes use of more specific and observable indicators. First, it considers whether the SoR includes a substantive explanation, whereby this is meant a minimal standard—namely, whether the explanation given identifies at least one specific reason for the restriction, for example a content category or a policy provision. A SoR that lacks such an explanation cannot be considered meaningful, but one that does include an explanation might still not meet the requirements of Article 17(4). Second, the analysis looks at vagueness by checking whether the platform makes its decision on the basis of general or residual categories rather than a more specific ground for moderation.

The analysis proceeds in two steps: first, descriptively: how often do the reported moderation decisions include substantive explanations, and how does this vary across platforms and the two transparency measures introduced above? Second, associatively: are differences in explanation provision linked to observable features of the moderation pipeline, such as automation, user-initiated reporting, or account termination?

The associative step is necessarily modest. We cannot observe a platform’s internal decision to explain or not explain a particular case nor we can establish its motivation. We can examine which specific features of the moderation pipeline are associated with an explanation provision. This pattern of associations lets us infer, indirectly, what kind of incentive drives explanation-giving. If, for example, more automated decisions are associated with higher explanation rates,

that would point to an infrastructural explanation: automation may produce structured outputs that are cheaper to convert into standard notifications. It would not establish that platforms are genuinely committed to transparency, only that explanation provision tracks the technical organization of the moderation pipeline.

Our theoretical expectation in this article is, therefore, not that platforms either comply or fail to comply in an uniform way.. Rather, we expect explanatory practices to reflect the relative costs and institutional incentives associated with producing justification through which Article is implemented at scale. Those costs should vary at three points in the moderation process: whether the decision is produced through automated systems that can generate structured outputs; whether enforcement is triggered by an external notice or by the platform itself; and whether the restriction is severe enough to make contestation more likely. Our analysis therefore focuses on three observable dimensions: the role of automated decision-making, the relationship between external notifications and platform enforcement, and the severity of the restriction imposed.

3.2 Data and case selection

The empirical analysis uses data from the DSA Transparency Database.¹ The database compiles detailed metadata on content-moderation actions reported by online platforms operating in the European Union under Article 24(5) DSA. Each record corresponds to one SoR and contains, inter alia, the date of the moderation decision, the type of content targeted, the DSA Annex I category invoked, whether the detection and the decision were automated, the type of notification that triggered the decision, the action taken (content removal, account restriction, account termination), and the textual explanation provided to the user, if any.

The analysis focuses on seven major dating applications operating in the EU: *Badoo*, *Bumble*, *Hinge*, *Meetic*, *Tinder*, *Lovoo* and *OKCupid*. Platform selection is motivated by three considerations: (i) dating platforms constitute a sector in which content moderation directly engages users' relational and identity interests; (ii) the seven platforms collectively vary in corporate structure, moderation technology, and reporting practices; and (iii) the homogeneity of the sector.

Rather than drawing a sample, the analysis uses the full population of SoR submitted by the seven platforms during a six-month observation window (1 December 2025 to 31 May 2026). Platform-level daily records are disaggregated to row level, yielding an analysis dataset of 6,878,730 observations. This full-population, bounded-window design avoids the imprecision of a stratified draw, but limits the analysis to six months rather than the multi-year window available in the underlying database.

The resulting platform sizes are highly unbalanced by construction (Figure 1), reflecting differences in platform scale rather than sampling choice. This imbalance does not bias the fixed-effects estimates, which remain consistent regardless of relative group size (Wooldridge, 1999), but it makes platform-specific estimates for smaller platforms (Hinge, OKCupid) less precise. The separate-platform model for LOVOO reported in Table 11 is also affected by a rank-deficient design matrix, as discussed below.

¹See <https://transparency.dsa.ec.europa.eu/>

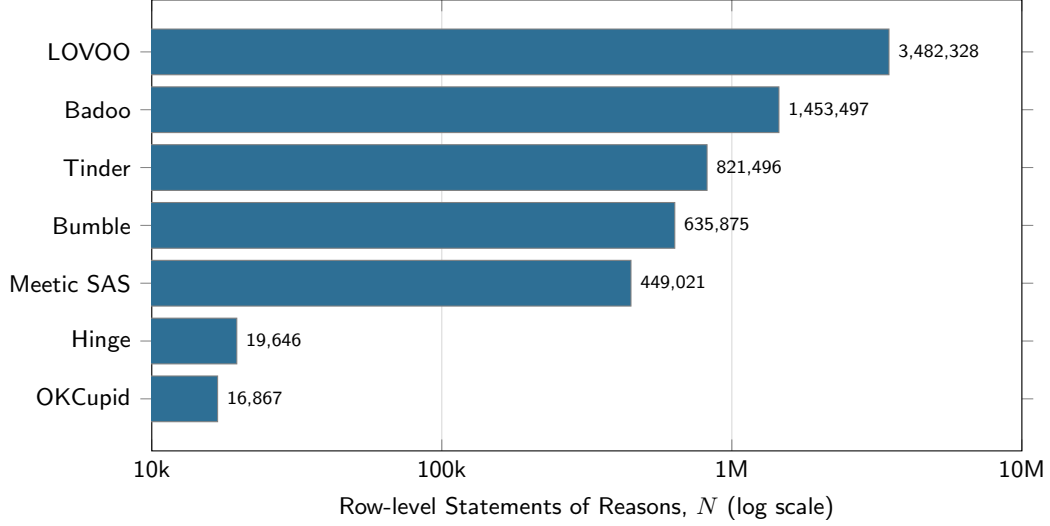


Figure 1: Number of Statement of Reasons (SoR) per platform in the EU DSA Transparency database
 Interpretation: Each bar shows the total number of SoRs per platform included in the analysis. The dataset covers the full population of SoRs submitted over a six-month observation window (1 December 2025–31 May 2026), not a sample. As shown, platform sizes are highly unbalanced, ranging from 16,867 (OKCupid) to 3,482,328 (LOVOO). The x-axis uses a logarithmic scale so that all seven bars remain comparably legible despite this range. Source: own elaboration, based on the DSA Transparency Database.

To test the three hypotheses developed in the previous section, the empirical analysis estimates a series of logistic regression models of the following form:

$$\begin{aligned}
 \text{logit}(\Pr(Y_{ip} = 1)) = & \alpha + \beta_1 \text{ContentType}_{ip} + \beta_2 \text{AccountTerminated}_{ip} \\
 & + \beta_3 \text{UserNotice}_{ip} + \beta_4 \text{AIDetection}_{ip} + \beta_5 \text{AIDecision}_{ip} \\
 & + \beta_6 (\text{AIDetection}_{ip} \times \text{AIDecision}_{ip}) + \delta_p + \varepsilon_{ip}
 \end{aligned} \tag{1}$$

M1 includes only *ContentType* and *AccountTerminated*; M2 adds *UserNotice*, *AIDetection* and *AIDecision*; M3 adds the platform fixed effects δ_p ; M4 adds the *AIDetection* $\times$ *AIDecision* interaction term (Table 7).

3.3 Operationalization of dependent variables

The dependent variables capture two distinct but complementary dimensions of Article 17 DSA disclosures. As the preceding doctrinal analysis showed, the transparency obligation imposed by Article 17 is not exhausted by the mere existence of a notification. Its value depends on whether the information communicated is specific and intelligible enough to help users understand the basis of the decision and potentially contest it. The Transparency Database does not allow us to assess that standard in full. It allow us, however, to observe two narrower dimensions: whether the user-facing explanation contains some substantive explanatory content; and whether the platform’s regulatory classification system when reporting moderation decisions to the DSA Transparency Database remains generic or residual.

These dimensions are conceptually separable. A platform may provide a relatively specific user-facing explanation while relying on highly generic reporting categories in the Transparency

Database. Conversely, it may select a relatively specific Annex II category while providing no substantive textual explanation to the affected user. To capture this dual structure, the analysis employs two complementary dependent variables: *Explanation Provided*, which measures user-facing explanatory specificity, and *Vague Explanation*, which measures opacity in the platform’s classification of the decision. Table 1 summarizes the operationalization.

Table 1: Operationalization of the dependent variables

Variable	Definition	Coded 1 when...	Coded 0 when...	Source field(s)
Explanation Provided	User-facing transparency: whether the affected user receives a substantive justification	The explanation field names a specific content category or policy provision	the field is empty/missing, or one of five boilerplate non-explanations (Table 2)	incompatible content explanation field
Ground vagueness	Specificity of the platform’s own internal infringement classification	The incompatible-content ground field is a generic/residual label (Table 3)	the ground field names a substantive, non-residual category	incompatible content ground field
Category vagueness	Specificity of the standardized DSA Annex I category	The decision selects one of the Commission’s residual Annex I categories (Table 3)	the Annex I category names a specific type of prohibited/illegal content	DSA Annex I category field
Vague Explanation (strict)	Reporting transparency: opacity on both classificatory levels at once	Ground vagueness = 1 AND Category vagueness = 1	either dimension is specific	derived from rows 2-3

All four variables are constructed from the raw DSA Transparency Database fields; see Section 4 text for the full coding rules.

The primary dependent variable, *Explanation Provided*, measures whether the platform communicated a substantive justification to the affected user. The variable is coded as 1 when the *incompatible content explanation* field contains any specific reference to the relevant content category or policy provision invoked by the platform, and 0 otherwise (Table 1). This variable operationalizes the user-facing dimension of Article 17 by capturing whether the recipient of the moderation decision received information capable of explaining the basis of the restriction.

The coding threshold is intentionally permissive. Any reference to a specific content category or policy rule is treated as constituting an explanation, even where the information provided is brief or highly standardized. Examples include *Lovoo* labels such as “Picture: Screenshot” or “Underage: Look”, as well as Bumble Inc. policy excerpts identifying a prohibited category of content. By contrast, only responses conveying no substantive information whatsoever are coded as 0. These include: (i) empty or missing values; (ii) the Match Group boilerplate statement “These content violations may not directly relate to trust and safety but still undermine the intended use of our platform as outlined in our Community Guidelines” (n = 203,449); (iii) “The account has been moderated for inappropriate behavior or sharing inappropriate content” (n = 5,223); (iv) “The account was moderated for sharing inappropriate content, which was removed from the account” (n = 1,714); and (v) “The account has been blocked for inappropriate

behavior or sharing inappropriate content” (n = 91).

Table 2: Categories coded as “no explanation” (Explanation Provided = 0)

Category	Example wording	n
Missing / empty	(no text in the explanation field)	—
Match Group boilerplate	“These content violations may not directly relate to trust and safety but still undermine the intended use of our platform as outlined in our Community Guidelines”	203,449
Generic “moderated” wording	“The account has been moderated for inappropriate behaviour or sharing inappropriate content”	5,223
Generic “removed” wording	“The account was moderated for sharing inappropriate content, which was removed from the account”	1,714
Generic “blocked” wording	“The account has been blocked for inappropriate behaviour or sharing inappropriate content”	91

The secondary dependent variable, *Vague Explanation*, captures a different dimension of disclosure under Article 17 DSA: the degree of specificity with which platforms classify moderation decisions within the DSA reporting framework. The variable is coded as 1 when the decision is simultaneously vague at two distinct classificatory levels, and 0 otherwise (see Table 3).

The first dimension, *ground vagueness*, is derived from the *incompatible content ground* field, which reflects the platform’s own internal classification of the infringement. A decision is classified as vague at this level when the platform relies on a generic or residual category lacking substantive normative content. This includes labels such as “*Non-T&S TOS Violation*” (Match Group), “*Other*” or “*Upgrade – Blocked by mistake*” (Bumble Inc.), “*Content violates terms and conditions*” (Lovoo), the raw DSA labels *decision ground illegal content* and *decision ground incompatible content* used as substantive grounds, as well as missing values.

The second dimension, *category vagueness*, is derived from the standardized Annex II category field. Unlike the previous measure, this field is selected from a fixed vocabulary established by the European Commission. A decision is classified as vague at this level when the platform selects one of the residual categories that fail to identify any specific form of harmful or prohibited content, namely generic terms-and-conditions violations or content deemed incompatible with the platform’s intended purpose.

The variable *Vague Explanation* equals 1 only when both dimensions are simultaneously vague. This constitutes the strictest operationalization of opacity adopted in the analysis, capturing moderation decisions that fail to provide meaningful specificity regardless of whether one examines the platform’s internal classification or the standardized DSA taxonomy.

Together, *Explanation Provided* and *Vague Explanation* capture two conceptually distinct aspects of Article 17 disclosures. *Explanation Provided* concerns the specificity of the user-facing reason communicated to the affected user. *Vague Explanation* concerns the specificity of the platform’s reporting classification in the DSA Transparency Database. Neither variable measures

Table 3: Vagueness classification labels, by dimension

Dimension	Residual / generic label	Platform(s)
Ground vagueness	“Non-T&S TOS Violation”	Match Group (Tinder, Hinge, OkCupid, Meetic)
Ground vagueness	“Other” / “Upgrade – Blocked by mistake”	Bumble Inc.
Ground vagueness	“Content violates terms and conditions”	Lovoo
Ground vagueness	raw DSA labels (“decision ground illegal content” / “incompatible content”) used with no further detail, or missing value	all platforms
Category vagueness	generic terms-and-conditions violation (residual Annex I category)	all platforms
Category vagueness	Content deemed incompatible with the platform’s intended purpose (residual Annex I category)	all platforms

meaningfulness in the full Article 17(4) sense; rather, they provide observable indicators of explanatory and classificatory specificity.

3.4 Explanatory variables and modelling approach

To examine the associative step of our strategy, we construct four process variables from the raw DSA Transparency Database fields and harmonize them across platforms. These variables corresponds to one of the three dimensions mentioned above automation, user-initiated reporting, and sanction severity. Each is designed to test a specific incentive-based prediction about explanation-giving, rather than to describe the moderated content itself.

Automation is captured with two variables. *AI detection* is coded as 1 when the automated detection field equals “Yes” and 0 otherwise. It captures whether algorithmic tools flagged content for review, regardless of how the ultimate moderation decision was made. *AI decision* is coded as 1 when automated decision equals “automated decision fully”, indicating that the moderation decision was made without human involvement, and 0 otherwise. Cases classified as “automated decision partially” (observed on *Badoo*, *Bumble*, *Hinge*, and *OKCupid*) are coded as 0 because they involve some degree of human review. Together, these variables are used to test H1 and allow the analysis to distinguish between non-automated, partially automated, and fully automated moderation pipelines.

User notice is coded as 1 when the moderation action was triggered by an external notification (i.e., *source type other notification* or *source article 16*) and 0 when the action was initiated by the platform itself (*source voluntary*). This variable captures whether explanation provision differs when enforcement begins with a user report rather than with platform-initiated detection and is used to test H2

Finally, *Account terminated* is coded as 1 when *decision account* equals “decision account terminated” and 0 otherwise. This latter category includes missing values, which are treated as indicating that no account termination occurred. The variable captures whether more severe

restrictions are associated with different explanation practices and is used to test H3.

Table 4: Construction of the process regressors

Variable	Coded 1 when...	Coded 0 when...	Notes
AI detection	automated detection field = “Yes”	otherwise	captures whether algorithmic tools flagged content, regardless of who made the final decision
AI decision	automated decision = “automated decision fully”	“Automated decision partially”, or a fully human decision	partial automation (observed on Badoo, Bumble, Hinge, OKCupid) is coded 0 because it involves human review
User notice	source type = “other notification” or “Article 16”	source type = “voluntary” (platform-initiated)	—
Account terminated	decision account = “decision account terminated”	Otherwise, including missing values	missing values are treated as “no termination occurred”

The regression models estimate whether these process variables are associated with the two dependent variables introduced above. The main specifications add explanatory blocks sequentially: first content type and sanction severity; then user notice and automation variables; then platform fixed effects; and, finally, an interaction between *AI detection* and *AI decision*. The interaction captures whether the association between automation and explanation provision depends on whether automation occurs at one stage of the pipeline or across both detection and decision-making. Section 4 first presents the descriptive patterns in the data, before turning to the regression results.

4 Results

4.1 Descriptive patterns: explanation provision and vagueness

The first step of the analysis is descriptive. It examines how often dating apps provide substantive explanations for the moderation decisions they report, and how this varies across platforms. Platforms explain, on average, 79.1% of the decisions they report to the Transparency Database, but this ranges from 7.1% to 98.9% depending on the platform, and a comparable share of decisions are classified using only generic regulatory categories (Table 4). Across the analytical sample, 79.1% of moderation decisions include a substantive explanation. However, substantial variation exists across platforms, ranging from 98.9% for *Hinge* and 96.0% for *Lovoo* to 7.9% for *Tinder* and only 7.1% for *Meetic* (Table 5).

The same table also shows the distribution of the strict vagueness measure. Across the sample, 53.6% of decisions satisfy the definition of *Vague Explanation*, meaning that they are vague both in the platform’s own ground classification and in the standardized Annex II category. Here too, the distribution is highly uneven: Badoo, Bumble, and Hinge rarely rely on vague classifications under this strict definition, while Meetic, Tinder, OkCupid, and Lovoo do

Table 5: Distribution of dependent variables by platform

Platform	N	Explanation provided (%)	Vague strict (%)
Badoo	1,453,497	95.4	0.1
Bumble	635,875	93.5	0.0
Hinge	19,646	98.9	1.1
Lovoo	3,482,328	96.0	71.6
Meetic	449,021	7.1	92.9
OKCupid	16,867	21.6	78.4
Tinder	821,496	7.9	92.1
Total	6,878,730	79.1	53.6

Explanation provided = 1: substantive explanation provided to the user. Vague strict = 1: vague decision at both ground and DSA Annex I category level simultaneously. N = 6,878,730 SoRs, 2025-12-01 to 2026-05-31

so far more often.

The two variables are related but not identical. *Explanation Provided* captures whether the user-facing explanation contains some specific explanatory content; *Vague Explanation* captures whether the platform’s reporting classification remains generic across both classificatory levels. Table 6 cross-tabulates the two measures:

Table 6: Cross-tabulation: Explanation Provided $\times$ Vague Explanation (pooled, N = 6,878,730)

Pattern	Share of decisions	Note
Transparent on both dimensions (explanation + specific category)	37.5%	—
Opaque on both dimensions (no explanation + vague category)	48.7%	—
Contradictory (inconsistent across dimensions)	13.8%	concentrated almost entirely in Lovoo, which writes specific content-type labels in the explanation field (e.g. “Picture: Screenshot”) but systematically selects generic Annex I categories
Total consistent (rows 1+2)	86.2%	—

The cross-tabulation shows that 86.2% of decisions are consistent across the two measures: 37.5% are transparent on both dimensions, combining a substantive explanation with specific regulatory categories, while 48.7% are opaque on both, combining no substantive explanation with vague classification. The remaining 13.8% are contradictory cases. These are concentrated almost entirely in Lovoo, which provides specific content-type labels in the explanation field, such as “Picture: Screenshot,” while systematically selecting generic Annex I categories. This dissociation reveals a third pattern of partial transparency, distinct from both full opacity and fuller observability.

Together with the explanation rate reported above, this completes the descriptive, first-step answer to our research question: platforms not only withhold a substantive explanation from the user in roughly one in five decisions, but they also default to a generic regulatory classification in just over half of all decisions. Figure 2 shows the distribution of SoRs by transparency pattern within each platform. The platform-level picture reinforces the unevenness visible in the three mutually exclusive categories introduced in Table 5.

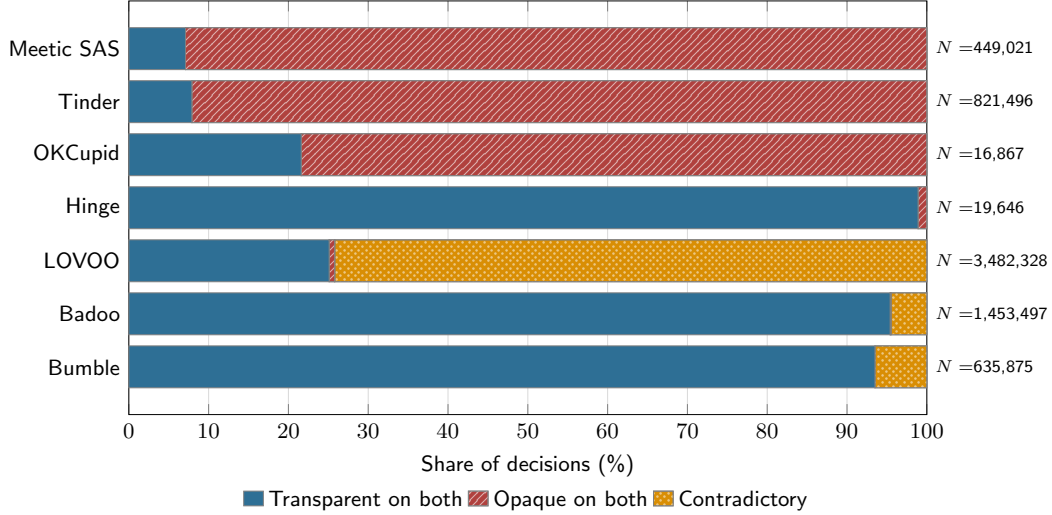


Figure 2: Level of transparency in Explanation Provided and Vague Explanation per platform.

Interpretation: Each bar shows the distribution of SoRs by transparency pattern within each platform, using the three mutually exclusive categories introduced in Table 5: transparent on both (a substantive explanation is provided and the regulatory category is specific), opaque on both (no explanation is provided and the regulatory category is vague), and contradictory (either a specific explanation is provided despite a vague regulatory category, or no explanation is provided despite a specific regulatory category). As shown, a clear pattern arises across platforms: three of them (Meetic SAS, Tinder, and OKCupid) show a high degree of opacity in their decisions; Bumble, Badoo, and Hinge, by contrast, provide a substantive explanation by default when content is moderated; and LOVOO stands out with a substantial share of SoRs classified as contradictory. Source: own elaboration.

These descriptive patterns show substantial variation in how explanations are provided and how specific reporting is. The next step is to examine whether that variation is associated with observable features of the moderation process.

4.2 Regression results

The second step of the analysis is associative. Building on the descriptive patterns reported above, this subsection examines whether variation in explanation provision and reporting specificity is associated with observable features of the moderation process: automation, the enforcement trigger, and the severity of the restriction imposed. The analysis proceeds in four steps, each adding a further block of explanatory factors to the previous one. This lets us see how much each block (the type of decision, the process that produced it, and the platform itself) contributes to explaining whether a platform gives a substantive explanation:

- M1.** Content type and decision severity (measured with the variable *account terminated*).
- M2.** M1 plus process characteristics (*User notice*, *AI detection*, *AI decision*).
- M3.** M2 plus platform fixed effects (a separate baseline for each platform, so that the model compares decisions within the same platform rather than across different platforms).
- M4.** M3 plus an $\text{AI detection} \times \text{AI decision}$ interaction term.

The interaction in M4 reflects how the structure of the moderation pipeline and how automation process works. AI detection and AI decision are not two independent features of a

platform, but two stages of the same pipeline: whether algorithmic tools flagged the content, and whether the final moderation decision was made without human involvement. When both stages are automated, the platform runs a fully automated pipeline that produces machine-readable output, which is easy to convert into a standard notification. When only one stage is automated (content flagged by an algorithm but reviewed by a person, or content flagged by a person but decided by an algorithm) the pipeline is incomplete, and there is no reason to expect the resulting notification to look the same. A model that simply adds the two variables together would assume that automating one stage always has the same effect regardless of what happens at the other stage, which is not a realistic assumption given how the pipeline works. We test this assumption with a likelihood-ratio test, a statistical test that compares how well the data fit the model with and without the interaction term: the test rejects the simpler, additive model in favor of the interaction model (Deviance = 93,597, 1 degree of freedom, $p < 0.001$).

Table 7 reports the four specifications for the primary dependent variable, *Explanation Provided*. Model fit improves steadily across the four models, with the log-likelihood increasing from $-2,051,865$ in M1 to $-1,076,681$ in M4. The largest improvement occurs when platform fixed effects are added in M3, indicating that platform-level differences account for a substantial part of the variation in explanation provision. The further improvement from M3 to M4 indicates that treating the two automation variables as simply additive is not supported by the data.

Table 7: Logistic regression — Dependent variable: explanation provided

	M1	M2	M3	M4
Intercept	2.355*** (0.001)	2.298*** (0.002)	2.944*** (0.008)	3.019*** (0.008)
Content type: text	-2.785*** (0.004)	-3.015*** (0.005)	0.370*** (0.009)	0.275*** (0.008)
Content type: other	-5.959*** (0.008)	-4.375*** (0.013)	-3.229*** (0.014)	-2.819*** (0.014)
Account terminated	3.232*** (0.012)	1.503*** (0.015)	1.088*** (0.013)	1.093*** (0.014)
User notice		0.355*** (0.008)	-1.120*** (0.008)	-1.118*** (0.008)
AI detection		-2.098*** (0.007)	0.357*** (0.009)	-1.493*** (0.010)
AI decision (fully automated)		2.408*** (0.008)	1.003*** (0.009)	-0.260*** (0.006)
AI detection x AI decision				3.430*** (0.011)
Platform FE	No	No	Yes	Yes
N	6,878,730	6,878,730	6,878,730	6,878,730
Log-Lik.	-2,051,865.2	-1,902,139.5	-1,123,479.3	-1,076,680.8

*** $p < 0.01$. Figures in parentheses are HC3 robust standard errors — a version of the standard error that stays valid even when the model’s prediction errors are more variable for some observations than others. Reference content category: image. M3-M4 include platform fixed effects (platform coefficients omitted for brevity). N = 6,878,730 across all models (row-level SoRs, 2025-12-01 to 2026-05-31, 7 platforms).

The move from M1 to M3 reflects a deliberate empirical strategy. In M1 and M2, the coefficients are estimated from two sources of variation at once: differences between decisions on the same platform, and differences between platforms. Relying on differences between platforms is risky, because platforms differ from one another in many ways we cannot observe, and any of those unobserved differences could be driving the result instead of the variable we are interested

in. We see direct evidence of it in the data: the coefficient on Content Type (text) flips from -3.02 in M2 to $+0.37$ in M3, and the coefficient on AI Detection flips from -2.10 to $+0.36$. In other words, before platform fixed effects are added, the model partly reflects platform-level differences: platforms that moderate a lot of text and rely heavily on algorithmic flagging, simply happen to explain less overall, for reasons that have nothing to do with text or algorithms as such.

Indeed, once we stop comparing across platforms, that spurious pattern disappears. Platform fixed effects are added in M3. In this sense, M3 and M4 no longer compare across platforms at all, it uses only differences between decisions made on the same platform and averages that kind of within-platform comparison across all seven platforms. This removes any bias coming from a platform’s fixed characteristics, whether we can observe them. To make the logistic-regression coefficients easier to interpret, we also report Average Marginal Effects (AMEs) for M3. AME translates a coefficient into the average change in the probability that an explanation is provided that is associated with a given variable, holding everything else constant (Table 9).

Table 8 reports the parallel four-model series for the secondary, reporting-transparency outcome, *Vague Explanation*. The same general lesson appears here: platform fixed effects substantially change the estimates, again indicating that platform-level reporting practices drive the observed patterns. M2 could not be estimated by standard maximum likelihood without platform fixed effects: image content, no account termination, a user-triggered notice, and a fully automated decision together almost perfectly predict a vague classification for Badoo and Bumble ($n=86,165$), a textbook case of quasi-complete separation. M2 is therefore estimated with Firth-type penalized likelihood, which resolves the separation and returns finite, stable coefficients.

As with *Explanation Provided*, adding platform fixed effects in M3 changes several coefficients sharply: *Content type*: text moves from $+1.92$ in M2 to -1.57 in M3, while *AI detection* shrinks from -2.67 to -0.31 . The interaction added in M4 is significant here too ($\Delta\text{deviance} = 37,924$, 1 degree of freedom, $p < 0.001$), and the pattern mirrors Table 7’s in reverse: automating only one stage of the pipeline is associated with a markedly more vague classification (AI detection alone: $+2.94$; AI decision alone: $+1.48$), while completing both stages largely, though not entirely, cancels this out (interaction = -4.02 ; combined effect $\approx +0.40$). Partial automation, in other words, is associated with worse reporting-transparency, not better, and only a fully automated pipeline approaches the baseline.

To make the *Vague Explanation* model easier to interpret, Table 9 reports average marginal effects for M3. These estimates express the association between each variable and the predicted probability of a vague classification in percentage-point terms. Account termination is associated with a substantially lower probability of vague classification, while user-triggered notices are also associated with less vagueness in the M3 specification. By contrast, content classified as “other” is associated with a higher probability of vague classification, while text-based decisions are associated with a lower probability compared with image-based decisions.

Taken together, the regression results show that explanation provision and reporting specificity are not evenly distributed across moderation decisions. They are associated with features of the moderation process, but they are also strongly shaped by platform-level differences. The

Table 8: Logistic regression — Dependent variable: Vague Explanation (reporting transparency)

Term	M1	M2	M3	M4
Intercept	-0.062***	0.641***	-9.427***	-8.500***
Content type: text	0.491***	1.920***	-1.568***	-1.132***
Content type: other	3.642***	8.154***	7.502***	5.134***
Account terminated	-4.461***	-7.874***	-2.349***	-2.251***
User notice	—	-2.588***	-1.102***	-1.293***
AI detection	—	-2.670***	-0.307***	2.937***
AI decision	—	0.063***	1.334***	1.484***
AI detection x AI decision	—	—	—	-4.017***
Platform FE	no	no	yes (ref.: Badoo)	yes (ref.: Badoo)
Log-likelihood	-4,327,765.7	-3,466,239.0	-2,204,277.2	-2,185,315.2
N	6,878,730	6,878,730	6,878,730	6,878,730

*** $p < 0.01$. HC3 robust standard errors for M1, M3 and M4 (as in Table 7). M2 cannot be estimated by standard maximum likelihood: without platform fixed effects, image content + no termination + a user-triggered notice + a fully automated decision is an almost perfect predictor of a vague classification for Badoo and Bumble ($n=86,165$) — quasi-complete separation. M2 is instead estimated by Firth-type penalized likelihood (brglm2), which resolves the separation; its standard errors come from the profile-likelihood approximation, not HC3. Higher values indicate a MORE vague, i.e. less specific, statement of reasons. LRT, M3 vs. M4: $\Delta\text{deviance} = 37,924$ (1 degree of freedom), $p < 0.001$.

Table 9: Average Marginal Effects — M3, Vague Explanation

Term	AME
Account terminated	-28.7 pp
AI decision	+11.4 pp
AI detection	-3.5 pp
Content type: other (vs. image)	+30.3 pp
Content type: text (vs. image)	-20.1 pp
User notice	-13.6 pp

next subsection therefore examines whether the associations observed in the pooled models hold consistently across platforms.

4.3 Platform heterogeneity

The pooled models indicate that platform identity is the dominant predictor of explanation provision. This is consistent with the descriptive patterns in Table 5 where explanation rates range from 90% for *Hinge*, *Lovoo*, *Badoo* and *Bumble*, *OKCupid* in 21.6%, while *Meetic* and *Tinder* do so in roughly 7% of decisions..

The next question is whether the associations estimated in the pooled models are stable across platforms. We examine this in two additional analyses: (i) a separate logistic model estimated for each platform; and (ii) a model that lets the effect of AI decision vary by platform (a platform $\times$ AI decision interaction).

To see whether the effect of automation is the same on every platform, we estimated a model that lets the effect of AI decision vary by platform (excluding *OKCupid*, where every decision has the same automation status, so no such effect can be estimated for it). A likelihood-ratio test confirms that it does ($\Delta\text{deviance} = 185,029$, $p < 0.001$): the effect of automation on explanation provision is not the same across platforms. The platform-specific AMEs of AI decision range from +22.6 percentage points (*Bumble*) and +15.6 (*Tinder*) down to -14.4 (*Meetic*), with a near-zero effect on *Hinge* (+1.1) and *Lovoo* (-0.1).

Table 10: Platform-specific conditional AME of AI decision (platform $\times$ AI decision interaction model)

Platform	Conditional AME of AI decision
Bumble	+22.6 pp
Tinder	+15.6 pp
Badoo	+11.9 pp
Hinge	+1.1 pp
LOVOO	-0.1 pp
Meetic	-14.4 pp
OKCupid	excluded (zero variance in AI decision)

Likelihood-ratio test comparing the additive model to the interaction model: $\Delta\text{deviance} = 185,029$ (5 degrees of freedom), $p < 0.001$.

The separate-platform models reported in Table 11 reinforce this picture. The relationship between the process variables and the extent to which explanations are provided varies greatly from platform to platform rather than adhering to one broad pattern throughout the industry. In these different models, the AI decision is strongly positive for Bumble (+49.2 pp), positive for Badoo (+12.4 pp) and for Tinder (+3.6 pp), null for Hinge (0.0 pp), and negative for Meetic (-14.6 pp). The case of Lovoo cannot be reliably interpreted in this specification since the model matrix is rank-deficient, which means that the data contain insufficient independent variation to estimate all the coefficients.

Meetic is a particular outlier, discussed fully below: despite routing 30.0% of decisions through fully automated pipelines, its overall explanation rate is only 7.1%, and fully automated decisions are associated with fewer explanations rather than more. The positive global AME of AI decision is driven primarily by *Bumble* and *Tinder* rather than by *Meetic*. This matters

Table 11: Platform-specific AMEs from separate logistic models per platform (DV: explanation provided)

Platform (N)	Account terminated	AI decision	AI detection	Content: other	Content: text	User notice
Badoo (1,453,497)	-32.3 pp	+12.4 pp	-0.6 pp	+3.7 pp	+0.6 pp	-2.2 pp
Bumble (635,875)	-19.8 pp	+49.2 pp	-9.9 pp	+5.9 pp	+5.1 pp	-0.5 pp
Hinge (19,646)	n/e	0.0 pp	-5.7 pp	n/e	n/e	+0.1 pp
Meetic (449,021)	+10.7 pp	-14.6 pp	-3.8 pp	-0.4 pp	+31.1 pp	+2.1 pp
OKCupid (16,867)	0.0 pp	n/e	+46.9 pp	+18.2 pp	+91.7 pp	+66.1 pp
Tinder (821,496)	+5.4 pp	+3.6 pp	+8.5 pp	-12.2 pp	-2.1 pp	-6.0 pp
LOVOO (3,482,328)	not reliable — rank-deficient model matrix					

n/e = term could not be estimated for that platform — either because the predictor does not vary on that platform (e.g. every OKCupid decision has the same AI-decision status) or because it perfectly predicts the outcome, a situation statistical software cannot assign a finite coefficient to (this affects account terminated and content type for Hinge). LOVOO’s separate-platform model returned what statisticians call a rank-deficient matrix — in plain terms, the data did not contain enough independent variation for the model to estimate every coefficient reliably — so its results are reported here as unreliable rather than silently left out.

because it shows that automation does not mechanically produce explanation. Its association with explanation provision depends on the platform infrastructure through which automation is implemented.

5 Discussion

5.1 Uneven reason-giving and the infrastructure of Article 17 DSA

The results raise four issues for how we should read platform compliance with Article 17 DSA. First, the finding that 20.9% of decisions across a large and diverse set of dating platforms come with no substantive explanation is, on its face, a real compliance concern under Article 17(3)(e) DSA, which requires platforms to state the legal or contractual basis for the restriction. Yet this shortfall is not spread evenly: it is concentrated almost entirely in three platforms (*Meetic*, *Tinder*, *OKCupid*) and tracks specific, identifiable features of how those platforms process decisions. This points to opacity as the product of a regulatory design that does not make platforms pay a real cost for failing to justify their decisions, rather than as an administrative accident. Platforms appear to have adapted rationally to the DSA’s procedural logic: reporting a high volume of decisions while spending as little effort as possible justifying them.

Second, the relationship between automation and transparency turns out to be non-linear rather than simply positive. When both stages of the pipeline are automated (algorithmic detection and a fully automated decision), the probability of an explanation rises: by 9.0 percentage points from adding automated decision-making on top of algorithmic detection, or by

19.2 points from adding algorithmic detection on top of an already-automated decision-making. But when only one stage is automated, the effect runs the other way: algorithmic detection without an automated decision goes with 10.2 points less explanation, and an automated decision without algorithmic detection with 1.3 points less. In other words, automating half of the pipeline is, if anything, worse for transparency than automating none of it.

Table 12: Non-linearity of automation effects: conditional AMEs by pipeline stage (M4)

Stage moving 0→1	Conditional on	AME
AI detection	AI decision = 0 (partial automation)	-10.2 pp
AI detection	AI decision = 1 (full pipeline)	+9.0 pp
AI decision	AI detection = 0 (partial automation)	-1.3 pp
AI decision	AI detection = 1 (full pipeline)	+19.2 pp

Only completing both stages of the pipeline (detection AND decision automated) is associated with a net positive effect on explanation provision; automating a single stage is associated with a net negative effect relative to no automation at all.

This confirms the mechanism behind Hypothesis 1: transparency is a by-product of completing the automation pipeline, not a sign that a platform has committed, as a matter of policy, to explaining itself. Only when both stages are automated does the system generate machine-readable output that gets converted, almost automatically, into a standardized notification. Automating only part of the pipeline breaks that conversion process without giving the platform any other reason to explain the decision.

The Meetic case provides the clearest evidence for this interpretation. Meetic routes 30.0% of its decisions through fully automated pipelines, yet its overall explanation rate is only 7.1%, and within Meetic, automated decisions are associated with fewer explanations than non-automated ones (−14.4 pp conditional AME). Meetic has automated part of the decision process without building the notification template infrastructure that would convert automation into transparency. This is precisely what the theory predicts: where the DSA does not price non-justification, platforms will rationally automate decisions without investing in the notification side of compliance.

The finding has direct implications for Article 17 design. Mandating human review of automated decisions does not guarantee more informative communications to affected users. According to the data, it tends to produce fewer. What matters is whether the moderation pipeline, whatever its degree of automation, generates structured outputs with specific policy references.

Third, user-initiated reports are associated with systematically less explanation, not more. The M4 estimate for user notice is −6.1 pp (negative, significant, and robust across all specifications). Unlike the automation variables, this estimate is stable between the additive specification (M3: −6.4 pp) and the interaction specification (M4: −6.1 pp), indicating that the user-notice penalty is not an artefact of conflating the trigger of moderation with the completeness of the automation pipeline: reports that originate from users are moderated less transparently regardless of how the resulting decision is processed.

Even at this magnitude, the finding confirms H2 and is the most difficult to square with a good-faith compliance interpretation. This is consistent with the procedural substitution mechanism: the platform treats receipt of the complaint as sufficient legitimation for the action,

substituting for independent justification directed at the person whose content or account is affected. The DSA does not adequately counteract this substitution effect. The reporting obligation under Article 24(5) runs to the Database, not to the quality of the explanation provided to the user.

Fourth, platform identity, rather than the characteristics of the decision itself, is what best predicts whether an explanation is given. The single most striking quantitative finding in the analysis is how much adding the seven platform fixed effects improves the model: the log-likelihood rises by about 778,660 units, from $-1,902,140$ to $-1,123,479$. In plain terms, simply knowing which platform decided predicts whether that decision comes with an explanation far better than knowing anything about the decision itself (how automated it was, what triggered it, or how severe it was). This is not a statistical curiosity: it is the strongest evidence in the paper that compliance with Article 17 DSA is, above all, an institutional choice, rather than a by-product of what is being moderated or how. Platforms appear to have arrived at their explanation rates through their own investment decisions that reflect a rational response to a regulatory framework that does not make non-justification costly. The three lowest-performing platforms (*Meetic* 7.1%, *Tinder* 7.9%, *OKCupid* 21.6%) are not failing at transparency by accident; they have built systems that structurally produce vague statements of reasons.

The interaction analysis confirms that the global AMEs conceal substantial cross-platform heterogeneity and that no single feature of the moderation process guarantees transparency independently of platform-level infrastructure choices. This heterogeneity underscores a fundamental limitation of the current regulatory design: a transparency obligation that leaves content and format to platform discretion will produce a wide distribution of outcomes, and voluntary standardization will be adopted only where it aligns with existing platform interests.

The findings acquire particular significance when read considering the normative framework set out. Dating applications are not mere commercial intermediaries. They constitute essential social infrastructure through which millions of users exercise participation rights that extend beyond ordinary consumer entitlements. When a platform decides to remove a user's content or terminate their account, it exercises a form of delegated regulatory authority over that user's access to a socially vital sphere. The right to know the reasons for such a decision is not a courtesy; it is a precondition for the meaningful exercise of whatever contestation and redress rights the DSA provides. Our results demonstrate that, across three of the seven platforms studied, this precondition is systematically unmet. The DSA has established the procedural architecture of a rights-based framework without ensuring that the substantive content of that framework reaches users in practice. This is not merely a compliance deficit: it is a structural failure to translate legal rights into lived experience for the users most likely to be affected by moderation decisions that go unexplained.

5.2 Corporate standardization and the limits of platform plurality

As touched upon in Section 2, Article 17(3)–(4) DSA requires statements of reasons to be "clear and specific" and to achieve "as much precision and specificity as the circumstances reasonably allow," without defining what clarity or specificity mean in practice, leaving providers of intermediary services, dating apps included, to translate that standard into their own wording,

categorization, and templates. Section 2 flagged the risk this creates in the abstract: that absent firmer interpretative parameters, statements of reasons could harden into procedural formality rather than substantive disclosure. The analysis in Sections 3 to 5 shows that this risk has materialized, what existing literature has found for statements of reasonsSoRs on other types of online platforms.

What predicts whether a user receives a substantive explanation is not what happened to them, but which dating app sent the notice: platform identity swamps every other factor in the model. The clearest illustration is a single line of text. The most common non-explanation in the dataset is not credited to any individual brand, but to Match Group, namely the corporate parent behind four of the seven platforms studied. Mapped onto ownership, the platform-level results resolve into a stark pattern: every measurable Match Group brand - be it Tinder, OkCupid, Hinge, or Meetic - sits at the bottom of the sample, while the two online platforms under different ownership, i.e., Lovoo and Badoo, sit roughly sixty points ahead at the top. Two alternative readings are worth ruling out first: on the one hand, Match Group's brands simply moderate different content less amenable to explanation; on the other hand, four product teams independently converged on similarly thin wording. However, neither survives the boilerplate finding, and the more 'parsimonious' reading is a single compliance decision, made once, and deployed across products marketed to users as distinct platforms.

This is what the risk of corporate capture looks like in practice (Kausche, 2025; Griffin, 2025). Because the DSA is prescriptive about what providers of intermediary services must achieve while remaining silent on how, it hands them the discretion to define their own compliance, and the data shows that discretion exercised not platform by platform, but by a small number of corporate actors filling it once. For an individual user, this means the choice between dating apps offers no real safeguard: someone banned without explanation on Tinder would very likely fare no better switching to Hinge or OkCupid, since the same corporate decision underlies all three. This matters more for dating apps than for a less concentrated sector: Match Group alone reportedly generates over half of global dating-app revenue [REF] and owns four of the seven platforms studied here. "Sector-level" observability, therefore, risks capturing one corporate compliance department's choices rather than a plural picture of how dating platforms handle Article 17 DSA.

A second pattern in the results points toward the same conclusion of corporate capture from a different angle. As also discussed in Section 2, providers primarily ground their moderation decisions in breaches of their own terms and conditions, applying that single contractual standard uniformly across the Union rather than engaging with national law. This may simply be efficient: establishing that content is unlawful requires engaging with the law of the relevant Member State, potentially multiplied across twenty-seven jurisdictions, whereas enforcing a provider's own house rules requires consulting one document, once. Yet whatever the motive, the effect is the same: the substantive standard governing what counts as legitimate content on dating apps becomes a text the provider itself authored, rather than the body of law the DSA reference to the legal category of illegal content was built to enforce online.

For dating-app users, this is not a mere technical distinction. Terms and conditions are known to define permissible nudity narrowly, and to crystallise heteronormative, cisgender, and

more broadly oppressive understandings of gender and sexuality, rather than the wider range of expression the law itself does not prohibit (Dvoskin & Kadri, 2025). Corporate capture of the moderation standard, combined with the opacity of the statements of reasonsSoRs through which that standard is applied, therefore creates a mechanism through which the design-based stigmatisation discussed in the Introduction could be reproduced, and even amplified, at the level of individual moderation decisions. Whether it actually is, however, remains an open question that this article’s own data cannot resolve as already noted, the Transparency Database records only the provider’s own conclusion, not the content or context underlying it, leaving no way to distinguish, from the disclosed information alone, a decision that removed genuinely unlawful material from one that merely enforced a narrower, oppressive standard of acceptable sexuality.

5.3 Formal transparency and partial observability

The doctrinal distinction developed in Section 2 supplies the criterion for identifying this pattern. Transparency washing, in this setting, is not non-compliance: it is compliance with Article 17(3) DSA that fails the functional test of Article 17(4) DSA. A boilerplate statement that names a residual category satisfies the enumeration of elements while conveying nothing the recipient could use to understand or contest the decision. In this sense, the gap between formal disclosure and meaningful transparency that Suzor et al. (2019) identified well before the Regulation entered into force, and which the DSA’s procedural architecture has reproduced rather than closed (Griffin, 2025). The Regulation thus makes visible what platforms report, but not what they decided: the volume of statements of reasons in the Transparency Database registers as compliance precisely because the metric tracks formal provision rather than the substantive intelligibility that Article 17(4) DSA requires. This is consistent with what has been found across platforms more generally: recurring deficiencies in the specificity and internal consistency of statements of reasons (Kaushal R et al, 2024) and post-DSA disclosures that still lack the depth and standardization required for effective scrutiny (Zornetta, 2025).

6 Conclusions and Future research

This article has provided the first sector-specific empirical analysis of compliance with Article 17 DSA’s obligation to provide reasons for content moderation decisions, focusing on seven major dating applications. Drawing on the full population of 6,878,730 Statements of Reasons from the DSA Transparency Database and employing logistic regression with platform fixed effects and a comprehensive battery of robustness checks, we find that:

Only 79.1% of moderation decisions are accompanied by a substantive explanation, with extreme variation across platforms (7.1% in *Meetic* to 98.9% in *Hinge*; Table 4). The relationship between automation and transparency is highly non-linear: completing both stages of the pipeline (algorithmic detection combined with fully automated decision) is associated with a +9.0 to +19.2 pp gain in explanation probability depending on the conditioning stage, but automating only one stage reduces explanation probability by −1.3 to −10.2 pp relative to no automation (Table 12). Transparency is a technical byproduct of pipeline completeness, not of automation

per se. User-initiated reports are associated with -6.1 pp less explanation. A finding inconsistent with any good-faith compliance model and consistent with procedural substitution: the external complaint provides cover that reduces the perceived obligation to justify the decision to the affected user. Account termination is associated with $+3.3$ pp more explanation, consistent with severity-driven disclosure pressure that partially offsets the general structural bias against justification. Platform identity is the dominant predictor of explanation provision: adding seven platform dummies increases model log-likelihood by $\approx 778,660$ units, more than all substantive regressors combined. This is the signature of incentive-driven institutional adaptation, not administrative error. The findings are robust to link function, modelling approach, platform-level clustering, and the treatment of potentially endogenous AI variables (See Annex I).

These results have implications for the evaluation and revision of the DSA’s transparency framework. The duty to provide reasons, as currently operationalized, does not produce systematic disclosure: it produces a binary split between platforms that have built transparency into their notification infrastructure and those that have not. This outcome is predicted by, and consistent with, the theoretical analysis developed above. The DSA creates strong incentives to report decisions in volume but does not adequately incentivize substantive justification; the rational response for cost-minimizing platforms is to comply procedurally while providing as little specific content as possible. The consequence, for users of platforms in the non-compliant cluster, is that an adverse moderation decision affecting access to a socially essential space arrives without explanation and without the information necessary to understand, challenge, or appeal it.

A more effective regulatory approach would be needed to correct this incentive structure directly. This could be achieved by establishing minimum content standards for Article 17 statements rather than leaving content specification to platform discretion. Additionally, the finding that automated decisions are associated with more informative explanations suggests that mandatory use of structured machine-readable templates for all notifications, regardless of whether the decision was automated, could significantly improve the informational quality of SoRs. Most fundamentally, the regulation must be redesigned to align the observable compliance metric (volume of SoRs reported) with the substantive objective (users receiving explanations adequate to support informed redress).

Although consistent results regarding the moderation of content on dating apps have been achieved, the structure of our data means that casual claims cannot be made. Our models show that the explanations given by platforms when moderating content are not randomly distributed. Some key features are associated with greater transparency, such as the use of AI throughout the entire moderation process. However, we cannot prove the causal mechanism that explains these dynamics. We propose three main hypotheses that could be explored in future research, and which align with our results.

Our results can be summarized as follows:

- firstly, that content moderation decisions processed through a fully automated pipeline, combining automated detection and fully automated decision-making, are associated with a higher probability of receiving a substantive explanation than decisions processed through partially automated or non-automated pipelines (R1).

- Secondly, content moderation decisions triggered by external notifications are associated with a lower probability of receiving a substantive explanation than decisions initiated by the platform itself (R2).
- Thirdly, content moderation decisions involving account termination will be associated with a higher probability of receiving a substantive explanation than less severe moderation decisions (R3).

We translate each of them into a specific mechanism that we may be exploring to explain why platforms provide or not a substantive explanation. Regarding R1, the proposed mechanism is not that automated systems are inherently more transparent in a normatively meaningful sense. Rather, full automation can generate structured, machine-readable outputs that are readily converted into standardized notification templates citing the policy category or rule invoked. In this sense, technical standardization may lower the marginal cost of producing an explanation. We therefore expect substantive explanations to be more frequent when automated detection and automated decision-making are combined in a complete pipeline than when automation is partial or absent.

Regarding R2, the proposed mechanism is one of the procedural substitutions. When an external notification provides the trigger for enforcement, the platform may treat the existence of that notification as sufficient procedural basis for action, reducing the organizational incentive to generate an additional, independently reasoned justification directed at the affected recipient. If this mechanism operates in practice, externally triggered decisions should be associated with lower rates of substantive explanation than platform-initiated decisions.

Finally, R3 reflects the expectation that more severe sanctions increase the expected costs of opacity. Account termination creates a greater likelihood of appeal, regulatory scrutiny, and reputational consequences than less severe interventions. These pressures may generate a stronger incentive to provide a specific explanation, partially offsetting the general organizational tendency towards standardization and minimal justification.

The three hypotheses are not logically independent. Each captures a different manifestation of the same underlying theoretical proposition: that compliance with Article 17 is shaped by the incentive structure surrounding the production of explanations, rather than by a uniform organizational commitment to substantive disclosure. Where justification is technically inexpensive to generate (R1), procedurally displaced by an external trigger (R2), or comparatively costly to withhold (R3), different transparency patterns should emerge even though the underlying incentive logic remains the same.

7 Limitations

The empirical strategy in Sections 3-5 answers a narrower, operationalized version of the research question posed in the Introduction, and it is worth being explicit about where that operationalization stops. The logistic regressions with platform fixed effects can establish, with reasonable confidence: (i) descriptive patterns in whether and how specifically dating apps report moderation decisions; (ii) which observable characteristics of a decision (automation

stage, trigger, severity, content type) are statistically associated with those patterns, net of platform identity; and (iii) that platform identity itself explains more of this variation than any substantive regressor, which is consistent with an institutional or corporate explanation for the gap. What the model cannot establish is the mechanism behind that platform-level gap: distinguishing deliberate cost-minimizing design from differential investment in notification infrastructure, or from unobserved differences in the underlying moderation caseload, requires evidence beyond what a fixed-effects logistic regression on DSA Transparency Database metadata can supply. Nor can the model speak to the substantive correctness or fairness of individual moderation decisions, or to the discriminatory effects flagged in Section 2: the dependent variable Explanation Provided is deliberately coded at a low, formal threshold (any reference to a specific category or policy provision, however brief), so that a decision can register as “transparent” under this operationalization while still falling short of the functional standard that Article 17(4) DSA and the transparency-washing critique below demand. The claims that follow (in particular, the corporate-capture reading of the Match Group pattern) are accordingly framed as the most parsimonious interpretation consistent with the data, not as a causally identified finding.

Notes

1. In this article the definition of content moderation used in the DSA is used. According to DSA’s article 3 content moderation means the activities, whether automated or not, undertaken by providers of intermediary services, that are aimed, in particular, at detecting, identifying and addressing illegal content or information incompatible with their terms and conditions, provided by recipients of the service, including measures taken that affect the availability, visibility, and accessibility of that illegal content or that information, such as demotion, demonetisation, disabling of access to, or removal thereof, or that affect the ability of the recipients of the service to provide that information, such as the termination or suspension of a recipient’s account.

2. Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market for Digital Services and amending Directive 2000/31/EC (Digital Services Act).

3. European Commission–DG CONNECT, *Digital Services Act Transparency Database*, Directorate-General for Communications Networks, Content and Technology, 2023. <https://doi.org/10.2906/134353607485211>

⁴ Because the analysis uses the full population of SoRs for the observation window rather than a sample, the resampling check reported in wp.tex (alternate random draws) does not apply here. Robustness is instead assessed through the battery reported in Table 13: multicollinearity diagnostics, Firth-type bias-corrected estimation, a probit specification, platform-clustered standard errors, and re-estimation without the (potentially endogenous) AI variables, all of which leave the substantive conclusions unchanged. A temporal robustness check (first half vs. second half of the six-month window), the population-design analogue of a resampling check, was planned but not completed in this run and remains a follow-up.

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